Development of a Metal-Based Variable Density Range Modulator for Proton FLASH Therapy

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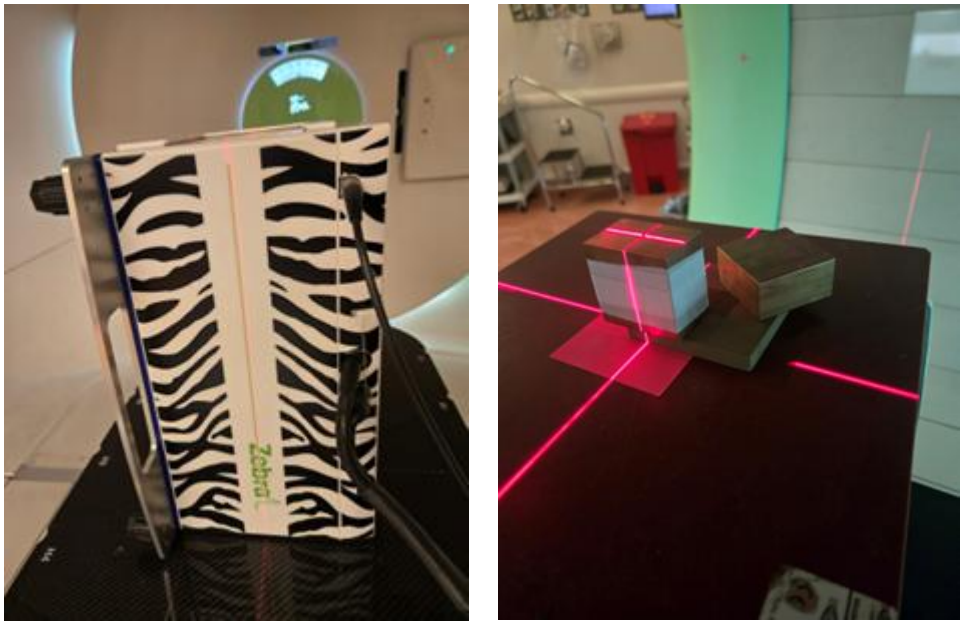
Purpose: Proton FLASH radiation therapy is an emerging modality with the potential to improve normal tissue sparing through ultra-high dose rate delivery. Achieving FLASH-compatible delivery with proton pencil beam scanning is challenged by energy layer switching times, which can exceed the temporal window associated with the FLASH effect. One proposed solution is the use of passive range-modulating devices inserted into monoenergetic proton beams to achieve depth-dose shaping without energy switching. While prior work has demonstrated plastic-based modulators capable of producing homogeneous depth-dose distributions, higher-density materials offer the potential for more compact devices with reduced material thickness. The purpose of this study is to experimentally characterize proton range modulation produced by metal-PLA composite materials using calibration phantoms, establishing the relationship between material infill and proton range shift to inform the development of metal-based variable density range modulators for proton FLASH applications.

Methods: Measurements performed at the Penn Medicine Roberts Proton Therapy Center using two metal-PLA calibration phantoms composed of brass and bronze with spatially varying infill to control effective material density. Each phantom consisted of nine discrete regions with manufacturer-specified infill ratios ranging from 40% to 100%, with infill defined as the fraction of material relative to air within each region. Variations in infill ratio were used to control the effective material density and corresponding proton stopping power. Identical experimental and analysis procedures were applied to both phantoms. Depth-dose measurements were acquired using a vertically oriented Zebra multilayer ion chamber (IBA Dosimetry) with 0.5 cm of solid water placed above the detector. The couch and Zebra were leveled prior to each measurement to ensure consistent vertical alignment relative to the beam axis (Figure 1). Beam delivery was defined using a brass block collimation system incorporating a central 2 x 2 mm slit when pressed together. Collimation performance was evaluated at proton beam energies of 115, 150, 180, 210, and 225 MeV. Transmission through the brass blocks was observed at energies ≥ 180 MeV. Therefore, all measurements were restricted to 115 and 150 MeV beam energies, for which effective collimation was confirmed. A single-spot pristine Bragg peak (PBP) was delivered with the gantry fixed at 0˚, the treatment snout fully retracted, and no range shifter inserted. For each phantom region, depth-dose curves were measured at the geometric center as well as at multiple off-center locations approximately 5 mm from region boundaries (Figure 2). An open-beam PBP reference was measured using identical experimental setup but without the calibration phantom in place. Depth-dose curves were analyzed in MATLAB to extract the distal 90% range (R_{90}) from normalized profiles. Range shift was quantified as

$$\Delta R_{90} = R_{90,PBP} - R_{90.region}$$

For each energy, regions were ranked by ΔR_{90} magnitude, and manufacturer-specified infill ratios were assigned according to this ranking prior to regression analysis. Linear regression was used to evaluate the relationship between range shift and infill (Figure 3).



(a) (b)

Figure 1. Vertical Zebra MLIC configuration showing (a) detector alignment using room lasers and (b) the measurement stack above the detector consisting of 0.5 cm solid water, a metal-PLA calibration phantom, and brass block collimation with a 2 x 2 mm slit.

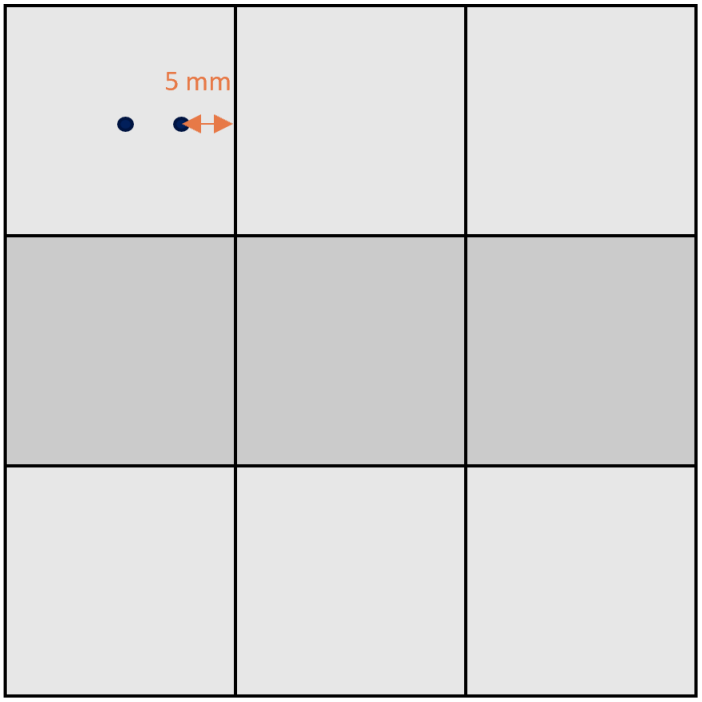


Figure 2. Center and off-center sampling locations within a representative phantom region. Depth-dose measurements were acquired at the geometric center of each region and at off-center locations approximately 5 mm from region boundaries.

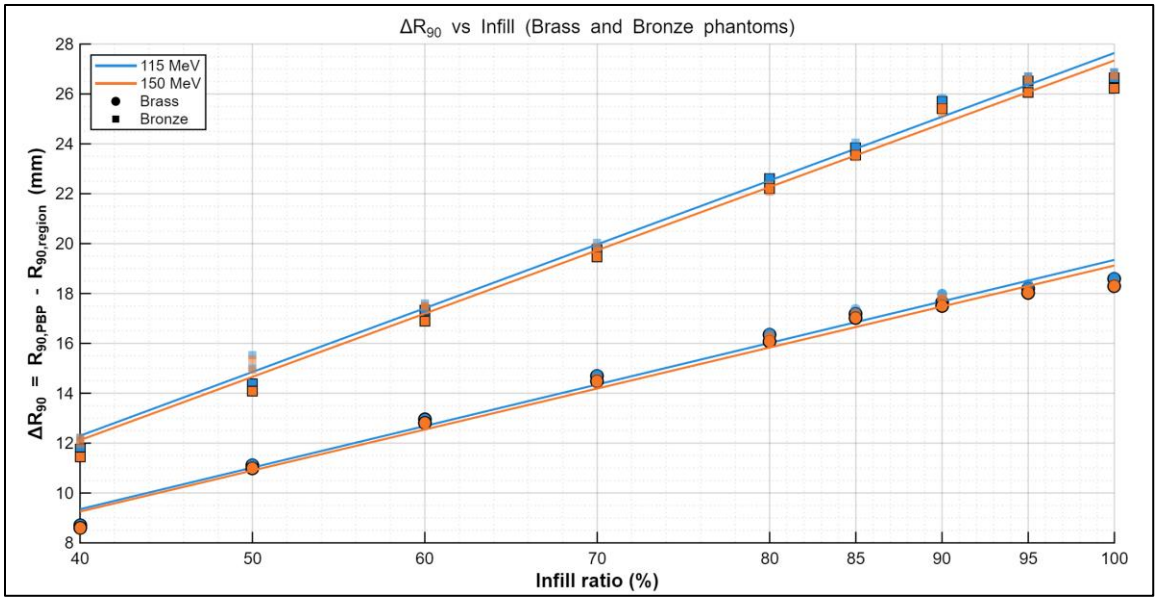


Figure 3. ΔR_{90} versus infill ratio for brass and bronze phantoms at 115 and 150 MeV. Filled and semi-transparent markers indicate on-center and off-center measurements, respectively. Solid lines represent unconstrained linear fits.

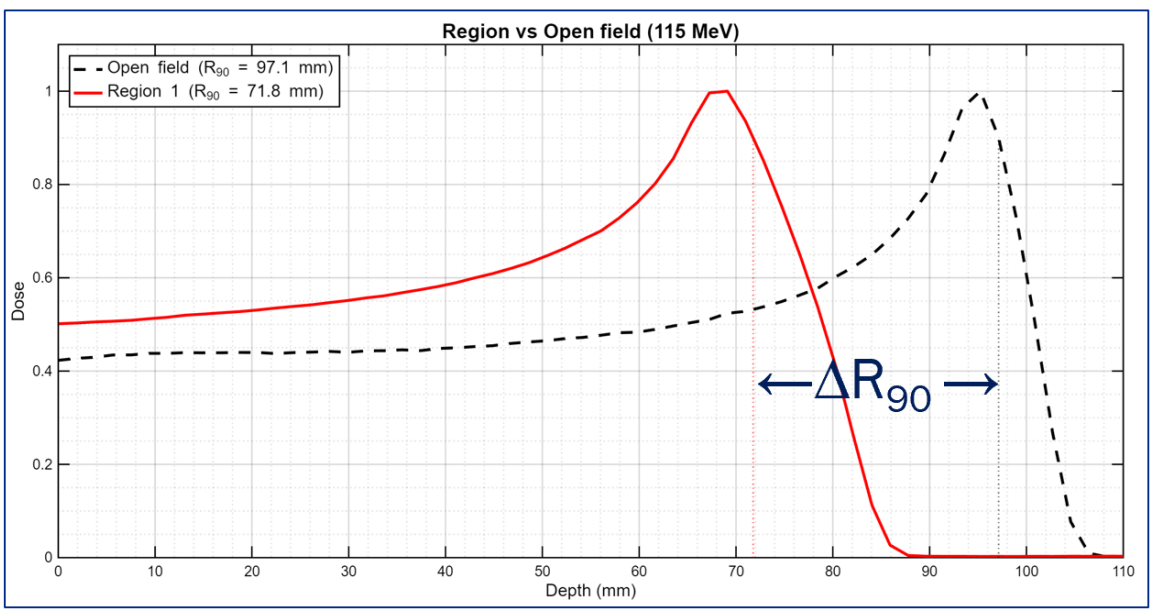


Figure 4. Example depth-dose comparison between open field and a representative phantom region at 115 MeV, illustrating the definition of ΔR_{90} as the difference in R_{90} between the two conditions.

Results: Depth-dose curves demonstrated clear distal range shifts for phantom regions relative to the open-beam PBP, enabling consistent identification of distal R_{90} values across energies and materials (Figure 4). For both the brass and bronze calibration phantoms, ΔR_{90} increased monotonically with infill ratio at 115 MeV and 150 MeV (Figure 3). Region ordering by ΔR_{90} was preserved across energies and between center and off-center measurement locations, with off-center measurements following the same ΔR_{90}-infill trend observed for center-only measurements. Comparable behavior was observed for both material phantoms.

Conclusions: This study demonstrated that infill within metal-PLA calibration phantoms produces predictable proton range shifts under fixed experimental conditions. For both brass and bronze phantoms, ΔR_{90} increased monotonically and approximately linearly with infill ratio at 115 MeV and 150 MeV, with consistent region ordering observed across energies and sampling locations. These findings support the use of infill as a controllable parameter for proton range modulation and directly inform the design of metal-based variable-density range modulators aimed at enabling depth-dose shaping from monoenergetic beams for proton FLASH therapy. Future work will focus on validating these experimental findings using Monte Carlo simulations performed in MATLAB and established proton transport frameworks.

Relevance to CIRMS: This work addresses a key measurement challenge relevant to proton FLASH therapy by experimentally validating infill-based proton range modulation in heterogeneous materials. The results support the development and verification of metal-based variable-density range modulators, contributing to CIRMS priorities in advancing measurement tools and methodologies for emerging proton therapy techniques.

References:

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2. U. Titt, M. Yang, X. Wang, et al., “Design and validation of a synchrotron proton beam line for FLASH radiotherapy preclinical research experiments,” Med. Phys. 49, 497–509 (2022).